

INTRINSIC FINITE-SHEET REDUCTION AND RECONSTRUCTION OF METRIC QUOTIENTS

QIAN QI

ABSTRACT. We give a reduction and reconstruction theory for metric quotients of polynomial observations. For a polynomial fast map with an isolated central fibre, explicit discriminant and derivative inequalities imply a finite-sheet normal form; the sheets are extracted from the original map rather than prescribed. The conclusion is uniform relative to the exact residual, including cancellation and ramified branch collisions, and determines an intrinsic tensor envelope. For an arbitrary number of invisible nonnegative coordinates, we reconstruct every minimal representation of the profiled quadratic form on a fully visible class. Its local Hessians form a Boolean lattice; they determine the minimal endpoint dimension and the normalized coupling matrix, leaving only positive coordinate scales and permutations. This class is nonempty and open in every endpoint dimension. A native categorical polynomial family has an exact rank and secant classification in every retained dimension, and its sharp fixed and adaptive ray complexities equal the dimension of a rank-one span. A finite-alphabet local experiment identifies these ray costs with efficient Gaussian discrimination, without identifying query counts with sample counts. Real local factorization and positive-time accessibility are proved separately.

1. INTRODUCTION

A singular inverse problem can lose information in three different operations: passage to a leading residual, minimization over coordinates, and removal of coordinates invisible at the target scale. These operations preserve more than an order of vanishing. They preserve a metric quotient, and the quotient can sometimes be reconstructed from finite intrinsic data.

The first question is when a finite affine description is forced by an observation map. Let

$$R(x, t) = (F(x) - a(t), G(x) - b(t)),$$

where $F : \mathbb{R}^r \rightarrow \mathbb{R}^r$ and $G : \mathbb{R}^r \rightarrow \mathbb{R}^q$ are polynomial. Theorem 2.2 constructs inverse sheets from the critical values of F and bounds their nonlinear distortion by rational derivatives of the original map. It does not assume independent monomial fast equations. Corollary 2.5 verifies the inequalities for coupled weighted initial maps, while Example 2.6 treats an arc approaching a genuine discriminant. The leading residual set and its tensor envelope are independent of the chosen inverse presentation.

The second question is whether a one-sided partial minimum remembers the quadratic form from which it came. Write

$$Q = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} > 0, \quad G_Q(x) = \min_{z \in \mathbb{R}_+^e} (x, z)^\top Q(x, z), \quad x \in \mathbb{R}_+^k. \quad (1.1)$$

Theorem 4.3 answers this question in every endpoint dimension on a fully visible class. The unlabelled local quadratic matrices of G_Q recover a Boolean lattice of active sets, the number e , and a unique normalized matrix up to permutation. The complete fibre among e -endpoint representations consists of positive diagonal congruences and permutations. No representation

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with fewer endpoints exists. A simple positive-spanning condition makes this an open, nonempty class for every e , not a list of isolated examples. Proposition 4.5 gives a finite coordinate-deletion criterion without the visibility hypothesis.

The third question concerns the native observation metric. Theorem 5.1 classifies the quotient image and all normalized secants of paired-contrast polynomial probability models. The answer is the linear span of the contrast tensors; no determinant witness is required. Corollary 5.2 places a native open family at every positive definite quotient, so the fully visible endpoint class and the finite-map metric reduction both occur in these genuine stochastic observations. Theorem 6.1 identifies the variational quotient with the nuisance-free Gaussian limit experiment. Its scalar values are intrinsic local discrimination costs. Recovering such values through an exact oracle remains a different problem from estimating them from samples.

These results have distinct, stated hypotheses. We do not identify every polynomial singularity with a finite affine arrangement, nor identify the paired-contrast class with every binary polynomial mixture. The reduction theorem supplies a converse for a broad finite-map class; the endpoint theorem supplies an all-dimensional reconstruction result; and the native theorem supplies an all-dimensional exactly classified stochastic class. The earlier binary model, its whole-model spectral limit, its rational-moment bounds, and its sharp two-variance theorem are retained in full in the supporting volume. The present proofs are independent of that volume except where an earlier application is explicitly identified.

Background and organization. Semialgebraic selection and one-variable preparation are used in the form developed in [5]; proper real uniformization and normal crossings are used in precisely the forms of [3, Theorem 0.1 and Corollary 4.9]. Convex separation, Schur complements, and quadratic optimization are classical [6]. The piecewise-quadratic active-set description of (1.1) is part of parametric quadratic programming [1], with its accompanying corrigendum [2]; our reconstruction concerns the unlabelled Hessian lattice, minimal representations, and the entire fixed-dimensional fibre. Comparison by randomizations goes back to Blackwell [4]; local asymptotic normality and efficient information are classical statistical constructions [7]. We provide their finite-alphabet proof here so that the relation to the singular root scale does not rest on a regularity analogy.

Sections 2–3 derive the finite quotient. Section 4 reconstructs its orthant profiling. Sections 5–6 treat native metrics and local experiments. Appendix A supplies the real-analytic factorization and accessibility arguments. Appendix B corrects a clock-pair quantifier and displays the partial fractions needed for the retained binary inverse.

2. REDUCTION FROM A POLYNOMIAL FINITE MAP

Let $N \subset \mathbb{R}^r$ be compact semialgebraic, the closure of its interior. Polynomials are defined on a neighbourhood of N . Suppose $F^{-1}(a_0) \cap N$ is finite, nonempty, and contained in the interior of N , and $G = b_0$ on that fibre. Let $a(t), b(t)$ be continuous semialgebraic arcs tending to a_0, b_0 . Put

$$\Delta = F(\{x \in N : \det DF(x) = 0\}) \cup F(\partial N). \quad (2.1)$$

Assume $a(t) \in F(N) \setminus \Delta$ for all sufficiently small positive t . Choose a positive continuous semialgebraic function $r(t) \rightarrow 0$ with

$$4r(t) < \text{dist}(a(t), \Delta).$$

When Δ is empty the distance restriction is omitted. The set $\Omega_t = \{x \in N : |F(x) - a(t)| \leq 2r(t)\}$ contains no critical or boundary point. Define the rational matrix and its fast derivative by

$$A_F(x) = DG(x)DF(x)^{-1}, \quad M(t) = \sup_{x \in \Omega_t} \|D_x A_F(x)DF(x)^{-1}\|. \quad (2.2)$$

The last norm is the operator norm of the derivative of a matrix-valued function. Also put

$$c(t) = \max\{|G(x) - b(t)| : x \in N, F(x) = a(t)\}.$$

All suprema are finite and semialgebraic. This follows from compactness, nonvanishing of $\det DF$ on Ω_t , and quantifier elimination.

Definition 2.1 (Polar-gap conditions). We say the marked residual satisfies the polar-gap conditions at radius r if

$$\sup_{\Omega_t} \|A_F\| \leq K_0, \quad c(t)/r(t) \longrightarrow 0, \quad \varepsilon(t) := r(t)M(t) \longrightarrow 0. \quad (2.3)$$

These are inequalities on the original polynomial map and the marked arc, not assumptions about the existence of a normal form.

Theorem 2.2 (Intrinsic finite-sheet reduction). *Under the preceding hypotheses and (2.3), the solutions of $F(x) = a(t)$ can be labelled by finitely many semialgebraic arcs $x_b(t)$, $1 \leq b \leq B$. There are inverse maps $\phi_b(U, t)$ on $|U| \leq r(t)$ with $F(\phi_b(U, t)) = a(t) + U$, and these maps exhaust the indicated fast tube. Set*

$$C_b(t) = G(x_b(t)) - b(t), \quad D_b(t) = A_F(x_b(t)).$$

Then

$$R(\phi_b(U, t), t) = (U, C_b(t) + D_b(t)U + E_b(U, t)), \quad |E_b(U, t)| \leq \frac{1}{2}M(t)|U|^2. \quad (2.4)$$

For H in a compact subset of Sym_{r+q}^{++} , let

$$q_b(H, t) = \min_{U \in \mathbb{R}^r} \|(U, C_b(t) + D_b(t)U)\|_H. \quad (2.5)$$

There is a uniform K such that, for all sufficiently small t ,

$$(1 - K\varepsilon(t)) \min_b q_b(H, t) \leq \min_{x \in N} \|R(x, t)\|_H \leq (1 + K\varepsilon(t)) \min_b q_b(H, t). \quad (2.6)$$

The distance is exactly zero if and only if some $C_b(t) = 0$.

If no C_b is eventually zero, write $C_b(t) = c_b t^{\rho_b} + o(t^{\rho_b})$, $c_b \neq 0$, and set $\rho = \max_b \rho_b$. The matrices $D_b(t)$ have limits D_b^0 , and the complete leading residual set is

$$\mathcal{E}_\rho = \bigcup_{\rho_b = \rho} \{(u, D_b^0 u + c_b) : u \in \mathbb{R}^r\}. \quad (2.7)$$

Here $\mathcal{E}_\rho$ is defined directly as the zero-time section of the closure of $\{(t, t^{-\rho} R(x, t)) : x \in N, t > 0\}$. In particular it does not depend on the inverse-sheet construction.

Proof. The restriction of F over $B(a(t), 2r(t))$ is a proper local diffeomorphism with finite nonempty fibres. Properness follows by taking inverse images inside the compact set N ; the absence of $F(\partial N)$ keeps them in its interior. A proper local diffeomorphism is a covering: disjoint inverse neighbourhoods of the finite fibre cover a sufficiently small target neighbourhood, since otherwise a sequence of additional inverse points has a compact limit in the fibre. The ball is simply connected, so every covering component maps diffeomorphically onto it.

Apply the same argument to the map $(t, x) \mapsto (t, F(x) - a(t))$ over $\{0 < t < t_0, |U| < 2r(t)\}$. After shortening t_0 , the arcs are analytic and the base is homeomorphic to an interval times a ball. The covering is therefore trivial. Its components and graphs are semialgebraic; alternatively their graphs are the finitely many definable continuations of the inverse fibre. This supplies the asserted labels and charts, including the closed smaller ball.

Differentiating on a sheet gives $D_U(G \circ \phi_b) = A_F \circ \phi_b$. Its derivative has norm at most $M(t)$ by (2.2). Taylor's integral formula on the convex U -ball proves (2.4).

For a direct relative comparison set $V = C_b + D_b U$. The first coordinate implies $|U| \leq |(U, V)|$, and compact metric bounds imply

$$|||(U, V + E_b)||_H - ||(U, V)||_H| \leq K\varepsilon(t)|||(U, V)||_H.$$

The constant is independent of C_b , including $C_b = 0$. The unrestricted minimizer in (2.5) is

$$U_b^* = -(L_b^\top H L_b)^{-1} L_b^\top H(0, C_b), \quad L_b = \begin{pmatrix} I_r \\ D_b \end{pmatrix}.$$

Since L_b contains I_r and D_b is uniformly bounded, $|U_b^*| \leq K'|C_b| = o(r)$. Thus this minimizer is a feasible trial. The trials $U = 0$ have cost $O(c(t)) = o(r(t))$. Outside the fast tube the norm of the first residual coordinate exceeds $r(t)$, so norm equivalence puts every global minimizer in a chart. Apply the preceding comparison there for the lower bound and at U_b^* for the upper bound. This proves (2.6). A zero forces $U = 0$, and then forces $C_b = 0$; the converse is immediate.

Every bounded semialgebraic matrix arc has a limit, and every nonzero semialgebraic vector arc has a nonzero rational-power leading vector. Uniform bounds on D_b show $q_b(H, t)$ is bounded above and below by positive constant multiples of $|C_b(t)|$: use the inverse of the bounded linear shear $(U, V) \mapsto (U, V + D_b U)$. Hence (2.6) selects the largest order ρ .

For completeness we identify the entire residual set, not only its minimum. A bounded normalized residual has $U = O(t^\rho) = o(r(t))$ and hence belongs to a chart. In that chart $t^{-\rho} E_b = o(1)$. A branch with $\rho_b < \rho$ cannot have a bounded normalized second coordinate because $D_b U = O(t^\rho)$. On a branch of order ρ every limit is $(u, D_b^0 u + c_b)$. Conversely choosing $U = t^\rho u$ realizes each such vector. These are the two inclusions in (2.7). $\square$

Corollary 2.3 (Coordinate invariance and uniformity). *Feasible source reparametrizations leave ρ and $\mathcal{E}_\rho$ unchanged. An invertible linear change T of residual coordinates sends $\mathcal{E}_\rho$ to $T\mathcal{E}_\rho$ and its tensor envelope to $TCT^\top$. The same conclusion holds for a smooth observation-coordinate change at the common limiting observation with invertible derivative T . Time is marked; replacing t by s^m multiplies the order by m .*

In a semialgebraic parameter family the estimate (2.6) is uniform whenever the metric bounds, K_0 , and the two limits in (2.3) are uniform. Formula (2.7) is uniform on compact strata on which the stated leading-vector and slope limits are uniform. No uniformity across a vanishing gap is implicit.

Proof. The first statements follow from the definition by the feasible residual image. For a smooth observation change, the difference of its values at two observations is $TR + o(|R|)$ uniformly as both observations approach the common limit, by the integral derivative formula. Every estimate in the proof of Theorem 2.2 uses only the displayed uniform constants and limits. This proves the final assertions. $\square$

The distinct affine spaces in (2.7), after duplicates are removed, are uniquely determined by the residual set: they have the same dimension r , and a real affine space cannot be covered by finitely many proper affine subspaces. This assertion concerns $\mathcal{E}_\rho$, not recovery of every sheet from the smaller tensor envelope.

Proposition 2.4 (Finite order tests). *For fixed semialgebraic data, the nonzero quantities in (2.3) and (2.2) admit Puiseux leading orders. Thus the two strict limits are equivalent to*

$$\text{ord } c > \text{ord } r, \quad \text{ord } r + \text{ord } M > 0,$$

with the convention $\text{ord } 0 = +\infty$. Boundedness of A_F is tested entrywise on the tube supremum. The branch residues c_b and limiting slopes D_b^0 , together with these order tests, extract the sheets. Scalar order data alone do not specify them.

Proof. Graphs of the suprema and of the finite inverse fibres are semialgebraic by quantifier elimination. One-variable semialgebraic preparation gives each nonzero leading power with nonzero coefficient. Taking the quotient or product gives the displayed equivalences. Apply the same preparation to C_b and D_b . Appendix A explains how proper real charts recover these orders without introducing inaccessible divisor faces. $\square$

Corollary 2.5 (Coupled weighted initial maps). *Give x_i positive rational weight w_i . Suppose*

$$F(x) = f(x) + F_{>1}(x), \quad f(t^{w_1}x_1, \dots, t^{w_r}x_r) = tf(x),$$

where every monomial of $F_{>1}$ has weighted degree greater than one and $f^{-1}(0) = \{0\}$. Let η be a regular value of f with a nonempty real fibre. Suppose $a(t) = t\eta + O(t^{1+\delta})$ for some $\delta > 0$, every monomial of G has weighted degree at least $\nu > 1$, and $b(t) = O(t^\nu)$. On a sufficiently small fixed neighbourhood of zero, Theorem 2.2 applies with radius $r(t) = c_0t$. In particular

$$c(t) = O(t^\nu), \quad \sup_{\Omega_t} \|A_F\| = O(t^{\nu-1}), \quad M(t) = O(t^{\nu-2}),$$

and the relative error is $O(t^{\nu-1})$. The fast initial equations may be coupled.

Proof. A weighted sphere is compact, and f is nonzero there. Weighted homogeneity therefore implies properness of f : its norm is bounded below by a positive constant times the weighted radius. Consequently the regular fibre at η is finite. A sufficiently small closed target ball about η avoids all critical values and is covered by finitely many smooth inverse charts.

Put $x_i = t^{w_i}\xi_i$. After division by t , the rescaled fast map converges in C^2 on bounded ξ sets to f . No inverse point over that target ball can escape to infinity in ξ . Indeed, on a small fixed weighted neighbourhood the higher-weight part of F is at most a small multiple of the lower bound for f , so $|F(x)| = O(t)$ implies bounded rescaled weighted radius. The inverse function theorem and compactness now give all inverse charts and uniform first and second derivatives in ξ . They also give a target ball of radius a fixed multiple of t avoiding critical and boundary values. Choose c_0 small enough to meet the factor four in the gap condition.

In a rescaled target coordinate $u = U/t$, the function G on these charts and its first two u derivatives are $O(t^\nu)$. One derivative with respect to U divides by t , and two divide by t^2 . These are exactly the estimates for A_F and M . The centre residual has the asserted bound as well. Thus $c(t)/r(t) = O(t^{\nu-1})$ and $r(t)M(t) = O(t^{\nu-1})$, proving the claim. Uniform versions follow on compact sets of the stated regular values, coefficients, and positive lower bounds. $\square$

Example 2.6 (Approaching a ramified discriminant). Take

$$F(x, y) = (x^2 + y^2, xy), \quad a(t) = (t, t/2 - t^{1+\lambda}), \quad \lambda > 0 \text{ rational}.$$

The real critical-value set is the union of the rays $a_1 = 2a_2 \geq 0$ and $a_1 = -2a_2 \geq 0$. For small positive t the four inverse points are given by

$$x + y = \sigma\sqrt{2t - 2t^{1+\lambda}}, \quad x - y = \tau\sqrt{2t^{1+\lambda}}, \quad \sigma, \tau \in \{-1, 1\}.$$

Thus the two inverse separations have different orders and the marked arc approaches a genuine critical-value wall. Choose $r(t) = c_0t^{1+\lambda}$ with a sufficiently small positive c_0 . If G is any vector-valued homogeneous polynomial of ordinary degree $n > 2 + 2\lambda$ and $b(t) = O(t^{n/2})$, then the polar-gap conditions hold. In fact, uniformly on the fast tube,

$$\begin{aligned} \|(DF)^{-1}\| &= O(t^{-(1+\lambda)/2}), & \|A_F\| &= O(t^{(n-2-\lambda)/2}), \\ M(t) &= O(t^{(n-4-3\lambda)/2}), & c(t)/r(t) &= O(t^{n/2-1-\lambda}). \end{aligned}$$

To check the second-derivative bound, differentiate $DG(DF)^{-1}$ once and multiply by $(DF)^{-1}$. The terms are bounded by $\|D^2G\|\|(DF)^{-1}\|^2$ and $\|DG\|\|D^2F\|\|(DF)^{-1}\|^3$; the latter gives the

displayed, weaker bound. Hence $r(t)M(t) = O(t^{(n-2-\lambda)/2}) \rightarrow 0$. Theorem 2.2 applies to all four competing sheets and every exact cancellation of their vector residuals. This example is not covered by requiring the target to remain a fixed relative distance from the rescaled discriminant.

3. THE TENSOR QUOTIENT EXTRACTED BY REDUCTION

For a closed residual set $E \subset \mathbb{R}^m$ define

$$\mathcal{C}(E) = \text{cl}(\text{conv}\{yy^\top : y \in E\} + \text{Sym}_m^+).$$

The order and the residual set in Theorem 2.2 give, for every $H > 0$,

$$\min_N \|R(x, t)\|_H = a_H t^\rho + o(t^\rho), \quad a_H^2 = \min_{y \in \mathcal{E}_\rho} y^\top H y. \quad (3.1)$$

The second equality follows either directly from (2.6) or by the two normalized-residual inclusions in its proof. We record the finite representation to make the reduction's output explicit.

Theorem 3.1 (Finite moment lift and metric faces). *Let $E = \bigcup_{b=1}^B (c_b + L_b \mathbb{R}^{r_b})$, where L_b has full column rank and $c_b \notin \text{ran } L_b$. Put $A_b = [L_b, c_b]$. Then*

$$\mathcal{C}(E) = \left\{ \sum_b A_b M_b A_b^\top + P : M_b \geq 0, P \geq 0, \sum_b (M_b)_{r_b+1, r_b+1} = 1 \right\}. \quad (3.2)$$

This set is closed without taking an additional closure. For $H > 0$, let $y_b(H)$ be the unique H -nearest point to zero on the b th sheet, and put $q_b(H) = y_b(H)^\top H y_b(H)$. The exposed face at the minimum of $S \mapsto \text{tr}(HS)$ is

$$\text{conv}\{y_b(H)y_b(H)^\top : q_b(H) = \min_c q_c(H)\}. \quad (3.3)$$

The envelope is recovered from its positive-metric costs by

$$\mathcal{C}(E) = \bigcap_{H>0} \{S \in \text{Sym}_m : \text{tr}(HS) \geq \min_b q_b(H)\}.$$

The active-sheet sets form a finite semialgebraic partition of the positive metric cone. Applied to (2.7), all these objects are invariants of the marked feasible observation germ.

Proof. The second moments of finitely supported affine random vectors give the identity

$$\text{conv}\{(u, 1)(u, 1)^\top : u \in \mathbb{R}^r\} = \{M \geq 0 : M_{r+1, r+1} = 1\}.$$

For clarity, if $M = \begin{pmatrix} K & m \\ m^\top & 1 \end{pmatrix}$, its covariance is $K - mm^\top \geq 0$. Diagonalize the covariance and realize each eigendirection by symmetric pairs about m ; the resulting finite distribution has second moment M . Conversely every second moment is positive semidefinite.

For each matrix in (3.2), let λ_b be its last diagonal entry. When $\lambda_b > 0$, divide by λ_b and use the identity just proved. When $\lambda_b = 0$, positivity makes its last row and column zero, so $A_b M_b A_b^\top$ may be absorbed into P . This proves equality before closure. Full column rank of A_b gives

$$\text{tr}(A_b M_b A_b^\top) \geq \sigma_{\min}(A_b)^2 \text{tr } M_b.$$

For a convergent sequence of represented matrices all positive summands therefore have bounded traces. Taking subsequences of their moment matrices and of P proves closedness.

Complete the square on each affine sheet. Equality in the minimum over (3.2) forces $P = 0$, zero covariance on every positive-weight sheet, and weight only on minimizing sheets, since H is positive definite. This gives (3.3). To recover the envelope, strictly separate an exterior matrix by a symmetric G . The positive-upper-set property forces $G \geq 0$. Replacing G by $G + \epsilon I$ preserves

the strict separation for sufficiently small $\epsilon > 0$, since the envelope consists of positive semidefinite matrices. The asserted intersection follows. Finally

$$y_b(H) = c_b - L_b(L_b^\top H L_b)^{-1} L_b^\top H c_b$$

shows that all comparisons of q_b are semialgebraic. The final invariance is Corollary 2.3. $\square$

The lift is a finite representation, not a claim of minimal semidefinite extension size. Nonzero ties of the leading costs do not determine which branch minimizes at finite time. The exact-zero assertion of Theorem 2.2, by contrast, is a finite-time equivalence.

4. RECONSTRUCTION AFTER INVISIBLE ENDPOINT ELIMINATION

Use (1.1). For $I \subseteq [e] = \{1, \dots, e\}$ put

$$\begin{aligned} z_I(x) &= -C_{II}^{-1} B_I^\top x, \quad z_{I^c}(x) = 0, \\ K_I &= \{x \geq 0 : z_I(x) \geq 0, B_j^\top x + C_{jI} z_I(x) \geq 0 \ (j \notin I)\}, \\ S_I &= A - B_I C_{II}^{-1} B_I^\top. \end{aligned} \tag{4.1}$$

Empty blocks have their natural meanings, in particular $S_\emptyset = A$. Strict positivity of C makes the endpoint minimizer unique. The stationarity and complementary inequalities show that the cones K_I cover the positive orthant and $G_Q(x) = x^\top S_I x$ on K_I . The formulas agree on overlaps. Thus G_Q is continuous and is quadratic on finitely many polyhedral regions. We use the quadratic coefficient S_I , namely one half the Hessian, as the label of a region.

Definition 4.1 (Full visibility). A representation is fully visible if B has column rank e and for every $I \subseteq [e]$ there is an $x \in (0, \infty)^k$ with

$$z_I(x) > 0, \quad B_j^\top x + C_{jI} z_I(x) > 0 \quad (j \notin I).$$

The first inequality has no entries when I is empty. These strict inequalities give an open region for each active set.

Let $\mathcal{H}(G)$ be the set of distinct symmetric matrices S such that $G(x) = x^\top S x$ on some nonempty open subset of $(0, \infty)^k$. The set, and the open regions carrying its labels, are determined by the function itself.

Lemma 4.2 (The visible Hessian lattice). *For a fully visible representation, $\mathcal{H}(G_Q) = \{S_I : I \subseteq [e]\}$ consists of exactly 2^e matrices, and*

$$S_I \geq S_J \iff I \subseteq J. \tag{4.2}$$

For $I \subseteq J$, $\text{rank}(S_I - S_J) = |J \setminus I|$. In particular A is the unique largest local matrix and $S_{[e]}$ the unique smallest one.

Proof. Set $P_I = A - S_I$. A block Schur complement, with $J = I \cup H$, gives

$$P_J - P_I = (B_H - B_I C_{II}^{-1} C_{IH})(C_{HH} - C_{HI} C_{II}^{-1} C_{IH})^{-1} (B_H - B_I C_{II}^{-1} C_{IH})^\top.$$

The middle matrix is positive definite and the outer matrix has column rank $|H|$, because the columns of B are independent. This proves the forward implication and its rank assertion. Conversely $P_I \leq P_J$ implies $\text{ran } P_I \subseteq \text{ran } P_J$: every vector annihilated by P_J has zero quadratic value for P_I and is annihilated by it as well. Since $\text{ran } P_I = \text{ran } B_I$, independence of the columns forces $I \subseteq J$. This proves (4.2) and distinctness. Full visibility makes all these polynomials occur on open sets. No additional polynomial can occur: any open set meets an interior cell of the finite polyhedral cover, and equality of quadratic polynomials on an open set implies equality of their matrices. $\square$

Theorem 4.3 (All-dimensional endpoint reconstruction). *Suppose $G = G_Q$ has a fully visible representation with $e \geq 1$ endpoints. The function G determines its minimal endpoint dimension, which is e , and every representation with exactly e endpoints as follows.*

Read $A = \max \mathcal{H}(G)$ and $S = \min \mathcal{H}(G)$. The e matrices covered by A in the ordered set (4.2) have rank-one deficits

$$A - S_{\{j\}} = u_j u_j^\top.$$

Choose x_0 in the open region with local matrix A , avoiding all cell boundaries, and orient each u_j by $u_j^\top x_0 > 0$. Put $U = (u_1, \dots, u_e)$ and $U^+ = (U^\top U)^{-1} U^\top$. Then

$$\bar{C} = [U^+(A - S)(U^+)^{\top}]^{-1}, \quad Q_{\text{norm}} = \begin{pmatrix} A & U \\ U^\top & \bar{C} \end{pmatrix}, \quad \text{diag } \bar{C} = \mathbf{1}. \quad (4.3)$$

These data are unique up to simultaneous permutation of the u_j and endpoint coordinates. A lexicographic ordering of the distinct oriented columns fixes a representative. All e -endpoint representations are its positive diagonal endpoint congruences and permutations. Their continuous fibre dimension is e .

The equivalent distance representations

$$G(x) = x^\top S x + \text{dist}(Tx, K)^2, \quad K = L\mathbb{R}_+^e, \quad (4.4)$$

of minimal dimension have the same S and the same pair (T, K) up to an orthogonal change of ambient coordinates. The normalized Gram matrix of the generating rays is $\bar{C}$.

Proof. In the original representation set $D_0 = \text{diag}(\sqrt{C_{11}}, \dots, \sqrt{C_{ee}})$. The rank-one deficits give $u_j = \pm B_j / \sqrt{C_{jj}}$. On the strict empty-active region all $B_j^\top x$ are positive, so the prescribed orientation selects the positive sign. This orientation does not depend on x_0 in that region. Thus $U = B D_0^{-1}$, up to permutation. The normalized coupling is $\bar{C} = D_0^{-1} C D_0^{-1}$ and

$$A - S = U \bar{C}^{-1} U^\top.$$

Multiplication by U^+ and its transpose gives (4.3). The normalized matrix is positive definite by congruence from Q . Its diagonal condition follows directly from the construction.

We next prove minimality without presuming visibility of an alternative representation. In any representation with e' endpoints, the range of every difference of two local quadratic matrices lies in the range of its cross block B' , by (4.1). The intrinsic difference $A - S$ has rank e . Hence $e \leq \text{rank } B' \leq e'$. No smaller representation is possible.

If $e' = e$, then B' has full column rank. The alternative representation has at most 2^e active-set polynomials, while G has exactly 2^e distinct local polynomials. Therefore every active set supplies a distinct full-dimensional cell. Each such cell has a strict point: a zero-coordinate or zero-multiplier equality is a nonzero linear equation because the columns of B' are independent, so finitely many such hyperplanes cannot cover an open cell. The alternative representation is fully visible. The preceding reconstruction applies to it and gives exactly the same normalized matrix, up to permutation. Thus every representation is obtained from (4.3) by

$$Q_D = \begin{pmatrix} A & UD \\ DU^\top & D\bar{C}D \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_e), \quad d_j > 0,$$

and a permutation. Conversely changing z by Dz preserves the nonnegative orthant and the infimum, so all these matrices represent G . The parameters d_j are freely and injectively specified by the endpoint diagonal. Permutations are discrete. This proves the fibre assertion.

For the distance statement, completing the square gives $S = A - BC^{-1}B^\top$, $C = L^\top L$ and $B = -T^\top L$. Conversely every such square expansion gives the positive matrix with $A = S + T^\top T$. Under the preceding congruence the generators become LD , up to permutation, which leave the cone unchanged. Any two invertible factors of a fixed Gram matrix differ by a left orthogonal

map, and $B = -T^\top L$ forces the same orthogonal map on T . Thus (T, K) has exactly the stated ambiguity. Normalizing the lengths of the rays gives $\bar{C}$. $\square$

Proposition 4.4 (An open visible class in every dimension). *Suppose B has column rank e and its kernel on the transpose contains a strictly positive vector:*

$$\text{there is } h \in (0, \infty)^k \text{ with } B^\top h = 0.$$

Then the representation is fully visible, for every $C > 0$ and every A with $A - BC^{-1}B^\top > 0$. These conditions describe an open nonempty class when $k \geq e + 1$. In particular one may take $k = e + 1$, let the first e rows of B form I_e and its last row be $-\mathbf{1}^\top$, choose any $C > 0$, and put $A = BC^{-1}B^\top + I_k$.

Proof. Surjectivity of $B^\top$ gives, for any $y \in \mathbb{R}^e$, a solution of $B^\top x = y$. Adding a sufficiently large multiple of h makes x strictly positive. Thus $B^\top(0, \infty)^k = \mathbb{R}^e$. For an active set I , choose $z_I > 0$, $z_{I^c} = 0$, $w_I = 0$, and $w_{I^c} > 0$. There exists an $x > 0$ with $B^\top x = -Cz + w$. These are precisely the strict stationarity and complementary inequalities, so I is visible.

The property persists under small perturbations of B . Indeed, solve the small residual in $B^\top h$ using a bounded right inverse of $B^\top$; the correcting vector tends to zero, so the corrected kernel vector remains positive. Rank, positive definiteness, and the Schur inequality are open conditions. The displayed matrix has positive kernel vector $\mathbf{1}$, and proves nonemptiness. Additional rows for $k > e + 1$ may initially be zero and the same positive-kernel argument applies. Thus this is an open family of genuinely coupled endpoint problems, not merely the diagonal case. $\square$

Proposition 4.5 (Finite redundancy criterion). *For an arbitrary positive definite Q , deleting endpoint coordinate j preserves G_Q if and only if, for every active cone $K_I^{(-j)}$ of the reduced endpoint problem,*

$$(B_j^\top - C_{jI}C_{II}^{-1}B_I^\top)x \geq 0 \quad \text{for every } x \in K_I^{(-j)}. \quad (4.5)$$

Each condition is membership of a specified linear functional in the dual of a polyhedral cone and can be tested by a finite linear feasibility problem. Independently of visibility, the invariant

$$\delta(G) = \dim \text{span}\{\text{ran}(S - S') : S, S' \in \mathcal{H}(G)\}$$

is a lower bound for every endpoint dimension. In the fully visible class it equals the minimal dimension e .

Proof. The reduced minimizer is feasible for the full problem with its j th coordinate zero. Its gradient in that coordinate, divided by two, is the left side of (4.5). The reduced stationarity conditions already hold in all other coordinates. The additional nonnegative gradient is therefore necessary and sufficient for full optimality. Strict convexity shows that equality of the two value functions for every x is equivalent to this condition at the unique reduced minimizer for every x . Substituting the finitely many active formulas gives (4.5), including boundaries. The range bound follows because every local matrix is $A - B_I C_{II}^{-1} B_I^\top$; every difference has range in $\text{ran } B$. In the visible class $A - S_{[e]}$ already has range of dimension e , giving equality. $\square$

The coordinate-deletion criterion does not assert that repeated coordinate deletion finds a minimal representation in every nonvisible problem. Theorem 4.3 proves global minimality and the complete minimal-dimensional fibre on its stated open class. Outside that class the finite equality test by intersections of active cones remains valid, but a full classification of all nonminimal representations is a separate question.

5. AN EXACTLY CLASSIFIED NATIVE STOCHASTIC FAMILY

Fix $z_1, \dots, z_m \in \mathbb{R}^k$ spanning $\mathbb{R}^k$. Consider a single categorical observation with $2m + 1$ cells and probabilities

$$p_{j,+}(a, \tau) = a_j/2 + z_j^\top \tau, \quad p_{j,-}(a, \tau) = a_j/2 - z_j^\top \tau, \quad p_0(a, \tau) = 1 - \sum_{j=1}^m a_j. \quad (5.1)$$

The centre is $\tau = 0$, where $a_j > 0$ and $\sum_j a_j < 1$. For all small τ in a fixed cone $\mathcal{K}$ with nonempty interior these probabilities are positive. The masses a are free nuisance coordinates and τ are retained coordinates. The observation metric is the ordinary, unaltered Hellinger metric. Thus the construction specifies the probability model and not an independently adjustable quadratic metric.

Theorem 5.1 (Native image, secants, and exact ray complexity). *For (5.1), profiling the free mass coordinates at the centre gives*

$$Q(a) = 4 \sum_{j=1}^m \frac{z_j z_j^\top}{a_j} > 0. \quad (5.2)$$

Put $\mathcal{V} = \text{span}\{z_j z_j^\top : 1 \leq j \leq m\}$ and $s = \dim \mathcal{V}$. The native image $\{Q(a) : a_j > 0, \sum a_j < 1\}$ is relatively open in $\mathcal{V}$, has dimension s , and its normalized nonzero secants are exactly the unit sphere in $\mathcal{V}$ for the Frobenius norm. In particular that secant set has dimension $s - 1$.

Exactly s legal ray values $v^\top Q v$, $v \in \mathcal{K}$, are necessary in the worst case to determine the unknown native quotient, for both fixed and deterministic adaptive exact queries. There is a fixed family of s rays attaining the bound. The same count determines the quadratic cone ball, and hence any fixed root decoder from which that ball's radial gauge is recovered.

Taking the contrasts e_i , $1 \leq i \leq k$, and $e_i + e_j$, $i < j$, gives $s = k(k + 1)/2$. Thus full-dimensional native families with sharp polarization complexity exist in every retained dimension, without varying the observation metric or using a numerical rank witness.

Proof. At $\tau = 0$, the two entries in pair j have probability $a_j/2$. Their retained probability derivatives are z_j and $-z_j$, while the mass derivative in that pair has entries $1/2, 1/2$. Their Fisher cross product cancels. The retained information is therefore $4z_j z_j^\top / a_j$, summed over pairs. The remaining cell has zero retained derivative. The nuisance information is

$$I_{aa} = \text{diag}(a_1^{-1}, \dots, a_m^{-1}) + \frac{\mathbf{1}\mathbf{1}^\top}{1 - \sum_j a_j} > 0.$$

Consequently its Schur complement is exactly (5.2). The spanning condition proves positive definiteness.

The coordinate change $a_j \mapsto c_j = 4/a_j$ maps the mass domain diffeomorphically onto an open subset of $\mathbb{R}^m$. The linear map $(c_j) \mapsto \sum_j c_j z_j z_j^\top$ is onto $\mathcal{V}$, and every surjective linear map is open. Hence the image is relatively open with dimension s . All secants lie in $\mathcal{V}$. Fix an interior image point and a sufficiently small relative ball around it. Its differences contain a relative ball about zero, so every nonzero direction in $\mathcal{V}$ is the direction of a secant. This proves the precise sphere assertion, including $s = 1$.

Evaluation by the rank-one forms $H \mapsto v^\top H v$ with v in the interior of $\mathcal{K}$ separates $\mathcal{V}$. Otherwise a nonzero quadratic polynomial would vanish on an open set. Finite-dimensional linear algebra therefore selects s such evaluations forming a basis of $\mathcal{V}^*$. Their values determine Q .

For the adaptive lower bound, fix Q_0 in a relative open ball of the native image and run a proposed algorithm making fewer than s queries. There is a nonzero $H \in \mathcal{V}$ annihilated by every recorded evaluation. Both $Q_0 + \epsilon H$ and $Q_0 - \epsilon H$ remain in the native image for sufficiently small

$\epsilon > 0$. They produce the same first answer as Q_0 , hence the same second query and answer, and inductively the same transcript and stopping decision. They are distinct, so the algorithm cannot recover both. No continuity of the query rule is assumed. A cone ball with nonempty interior determines its positive quadratic form by its radial gauge, proving the fibre assertion.

Finally the listed contrast tensors form a basis of Sym_k : the $e_i e_i^\top$ give the diagonal entries and subtracting two such diagonal tensors from $(e_i + e_j)(e_i + e_j)^\top$ gives the (i, j) off-diagonal tensor. This proves the all-dimensional statement. $\square$

Corollary 5.2 (Native realization at every positive quotient). *For every $Q_0 \in \text{Sym}_n^{++}$ there is a fixed paired-contrast experiment whose native quotient image contains a neighbourhood of Q_0 in Sym_n . Its design and contrasts are fixed, and only its centre masses vary. Consequently the fully visible endpoint class of Section 4 occurs in a native polynomial probability model with the ordinary Hellinger metric.*

Likewise, every finite-map reduction in Theorem 2.2 can be embedded, on a sufficiently small feasible neighbourhood of its central fibre, in a strictly positive polynomial probability observation. Given any $H > 0$ on its residual space, that embedding may be chosen so that its minimum Hellinger distance has the two-sided relative comparison

$$\min_x h(P_{\text{rem}}(x), P_{\text{base}}(t)) = \frac{1}{2} \min_b q_b(H, t)(1 + O(\varepsilon(t) + \omega(t))),$$

where $\omega(t) \rightarrow 0$ and equality here denotes the corresponding two-sided bounds, including exact zeros. Thus the extracted finite sheets and the positive-metric envelope are native stochastic objects for this class.

Proof. Choose vectors $v_1, \dots, v_N$ whose rank-one tensors form a basis of Sym_n , for example the vectors used in Theorem 5.1. Choose $\eta > 0$ so small that $Q_0 - \eta \sum_j v_j v_j^\top > 0$, and factor this difference as $WW^\top$ with columns $w_1, \dots, w_n$. Put $m = N + n$ and choose all centre masses $a_j = 1/(2m)$. Use contrasts

$$z_j = \sqrt{a_j \eta / 4} v_j \quad (j \leq N), \quad z_{N+i} = \sqrt{a_{N+i} / 4} w_i \quad (i \leq n).$$

Formula (5.2) gives $Q(a) = Q_0$. The contrast tensors span Sym_n , so the image is open there. Apply this construction with $n = k + e$ to any fully visible matrix Q_0 , marking its last e retained coordinates as nonnegative endpoints. Profile the free masses first and then those endpoints. This gives exactly G_{Q_0} , and all nearby matrices remain in the same native experiment.

For the reduction, write $O(x) = (F(x), G(x))$, $o(t) = (a(t), b(t))$, and $o_0 = (a_0, b_0)$. Apply the preceding construction to H and set the paired model's retained coordinate to $O(x) - o_0$ or to $o(t) - o_0$, with the same fixed masses. Every probability is polynomial in x on the remote side, and the marked base is semialgebraic. Positivity holds after restricting to sufficiently small neighbourhoods of the finitely many central inverse points. This restriction does not change the small-time minimizing germ, since the residual is bounded away from zero on the discarded compact part.

On a compact positive probability neighbourhood the square-root identity gives a uniform norm equivalence and, when both observations tend to the centre, the stronger relative comparison

$$h(P', P)^2 = \frac{1}{4} R^\top H R (1 + O(\omega)), \quad R = O(x) - o(t),$$

where ω bounds the two displacements from o_0 . Trials at the inverse origins have residual $O(c(t))$. Norm equivalence therefore localizes Hellinger minimizers and quadratic trials to the same fast tube and bounds their residuals by $O(c(t))$. At those points one may take $\omega(t) = K(|o(t) - o_0| + c(t)) \rightarrow 0$. Combining with (2.6) proves the asserted relative bounds, without dividing by a residual that may vanish. The contrasts span the observation space, so equality of the probability vectors is exactly equality of the observations. Exact cancellations are preserved. $\square$

Corollary 5.3 (Rank loci and fixed root patterns). *For a varying contrast matrix Z , the native dimension is exactly the rank of the matrix with columns $\text{vec}_{\text{sym}}(z_j z_j^\top)$. Its rank-drop loci are given by the corresponding minors. Positive mass changes cause no additional dimension drop. Under a prescribed retained-coordinate map $\tau = B\xi$, the formula is the same with contrasts $B^\top z_j$.*

Every fixed combination of simple root shifts, inward endpoint sums, and repeated interior variances can be used as the coordinates τ in (5.1). This gives polynomial probability observations with the same quotient classification on that pattern, and uniform conclusions on compact sets with positive cell margins and fixed contrast-rank bounds.

Proof. The dimension assertion is Theorem 5.1; polynomial minors give exactly the rank loci. Substitution gives the pullback contrasts. A simple root displacement and an endpoint sum are linear polynomial functions of root coordinates. For a repeated interior cluster the function

$$X = \frac{1}{2} \sum_{i=1}^d \left(u_i - \frac{1}{d} \sum_h u_h \right)^2$$

is polynomial, nonnegative, and realizes every small nonnegative value with zero mean. These choices independently realize the indicated product cone near separated clusters; the means of repeated interior clusters are held fixed, and bounded mass changes remain feasible. Substituting these polynomials into (5.1) proves the model assertion. Compact strict margins give the stated uniformity. $\square$

For two roots at $r \pm h$, the variance is $X = h^2$. In the coordinate $h \neq 0$ the pulled-back information is $4h^2 Q_X$, which vanishes as $h \rightarrow 0$. At the collision the nonzero variance coordinate is second order instead, and the local root scale is $n^{-1/4}$ rather than $n^{-1/2}$; this follows precisely from the local experiment below. Thus the rank formula distinguishes a coordinate transition from loss of the quadratic variance experiment. We do not extend compact fixed-pattern uniform constants across this degeneracy.

The class (5.1) is distinct from the two-component binary polynomial model. The latter retains its earlier rational-moment description and its proved sharp two-variance native family. The present theorem supplies a general classified native class; it does not assert that every higher-degree binary pattern has full dimension. Also, the unknown a is not supplied as part of an exact-query problem. If the full numerical centre law is supplied, the masses and (5.2) are directly known and no oracle reconstruction is needed.

6. THE LOCAL EXPERIMENT REPRESENTED BY A RAY COST

We use squared Hellinger distance $h^2(p, q) = \sum_e (\sqrt{p_e} - \sqrt{q_e})^2$, with no factor of one half. Let p be a strictly positive finite probability vector. Suppose the feasible local alternatives, indexed by $a \in \mathbb{R}^r$ and v in a cone $\mathcal{K} \subset \mathbb{R}^k$ with nonempty interior, have

$$p_{n,a,v} = p + n^{-1/2}(La + Dv) + o(n^{-1/2}), \quad (6.1)$$

uniformly on bounded index sets. The columns have zero sums and (L, D) is injective. The remainder is in ordinary Euclidean norm on the finite cell space. Define

$$W = \text{diag}(p_e^{-1}), \quad I = (L, D)^\top W (L, D), \quad Q = I_{DD} - I_{DL} I_{LL}^{-1} I_{LD}.$$

The empty free block has its natural convention. Fixed clock designs with limiting proportions $w_j > 0$ give the same formula with $W = \text{diag}(w_j/p_{je})$.

Theorem 6.1 (Canonical efficient discrimination). *For independent observations from (6.1), the local log likelihood has the expansion, for each bounded local index and uniformly on compact*

index sets when the remainders are uniform,

$$\log \frac{dP_{n,a,v}^{\otimes n}}{dP_{n,0,0}^{\otimes n}} = (a, v)^\top Z_n - \frac{1}{2}(a, v)^\top I(a, v) + o_P(1), \quad Z_n \implies N(0, I), \quad (6.2)$$

provided the centre is taken to be exactly $p_{n,0,0} = p$. After removal of the unrestricted nuisance translation, the Gaussian limit experiment is

$$Y \sim N(Qv, Q), \quad v \in \mathcal{K}. \quad (6.3)$$

Its scalar discrimination cost is

$$c(v) = v^\top Qv, \quad \text{Aff}(N(0, Q), N(Qv, Q)) = e^{-c(v)/8}, \quad \text{Power}_\alpha(v) = \Phi(\Phi^{-1}(\alpha) + \sqrt{c(v)}). \quad (6.4)$$

Here the last expression is the Neyman–Pearson power for testing the null against the specified efficient local alternative at level α .

Two such labelled quotient experiments with the same retained coordinates are statistically equivalent by Markov randomizations in both directions if and only if their matrices Q agree. Thus the ray costs determine an intrinsic discrimination object, not just a chosen matrix representation. Exact-ray complexity is nevertheless not sample complexity.

Proof. Write $s = La + Dv$ and $\delta_n = p_{n,a,v} - p$. Positivity gives a uniform bound on $|\delta_{ne}/p_e|$. Taylor expansion of $\log(1 + u)$, with the third-order remainder bounded by a constant times $|u|^3$, yields

$$\sum_{i=1}^n \log \frac{p_{n,a,v}(X_i)}{p(X_i)} = \sum_e (N_e - np_e) \frac{\delta_{ne}}{p_e} - \frac{n}{2} \sum_e \frac{\delta_{ne}^2}{p_e} + o_P(1).$$

The first-order expectation vanishes because $\sum_e \delta_{ne} = 0$. The replacement of the quadratic empirical counts by their expectations has order $o_P(1)$, and $n \sum_e |\delta_{ne}|^3 = O(n^{-1/2})$. The finite-dimensional multinomial central limit theorem and (6.1) now give (6.2). Uniformity follows from bounding the remainder against the norm of the same finite-dimensional count vector. The zero-sum score columns remove the $pp^\top$ covariance term. For a deterministic clock allocation, apply the same argument independently to each group and use $n_j/n \rightarrow w_j$.

The Gaussian score $Z = (Z_L, Z_D)$ has mean $I(a, v)$ and covariance I under the local alternative. The efficient score

$$Y = Z_D - I_{DL} I_{LL}^{-1} Z_L$$

has mean Qv , covariance Q , and zero covariance with Z_L . The two blocks are therefore independent. The mean of the nuisance block is $I_{LL}a + I_{LD}v$ and can take every value as a ranges freely. Equivalently, after the nuisance reparametrization $a \mapsto a + I_{LL}^{-1} I_{LD}v$, the Gaussian experiment is the product of the efficient experiment and an unrelated full translation family. Discarding that nuisance translation yields precisely (6.3). This is the quotient operation; it does not claim that data concerning an unknown nuisance parameter can be simulated without a parameter.

Whitening by $Q^{-1/2}$ gives normal distributions with common identity covariance and mean difference $Q^{1/2}v$. Integrating the product of square-root densities by completing the square gives the affinity in (6.4). The log likelihood has variance $c(v)$ and means $-c(v)/2$ and $c(v)/2$ under the null and alternative; its threshold test gives the displayed power, also at $c(v) = 0$ by continuity.

Identical Q give identical quotient experiments. Conversely Hellinger affinity cannot decrease under a Markov kernel: apply the Cauchy–Schwarz inequality conditionally on each output. Kernels in both directions therefore force equality of affinities of each labelled parameter pair, in particular of 0 and v . Formula (6.4) implies equality of $v^\top Qv$ on the cone. Since that cone has nonempty interior, the two quadratic polynomials and their matrices are equal. This proves the converse. $\square$

Corollary 6.2 (Variational costs and singular root scaling). *Suppose the feasible local chart has the stronger score expansion $p_{t,a,v} = p + t(La + Dv) + O(t^{1+\epsilon})$, $\epsilon > 0$, with bounded free coordinates realizing each bounded feasible v , and every almost-minimizing local law has bounded scaled coordinates. Then*

$$\lim_{t \downarrow 0} \frac{4}{t^2} \inf_a h^2(p_{t,a,v}, p) = v^\top Qv.$$

With $t = n^{-1/2}$ the same matrix is the efficient information in Theorem 6.1. In a repeated interior cluster, variance tv is realized by root displacements of order $n^{-1/4}$; inward endpoint sums and simple root shifts have order $n^{-1/2}$.

Proof. The square-root identity gives $4t^{-2}h^2(p_{t,a,v}, p) = \|La + Dv\|_W^2 + o(1)$ uniformly on bounded sets. Its minimum over a is $v^\top Qv$, obtained by $a_* = -I_{LL}^{-1}I_{LD}v$. Testing its feasible realization proves the upper bound; the bounded-coordinate assumption and the same expansion prove the lower bound for every almost-minimizing sequence. Setting $t = n^{-1/2}$ gives (6.1). The cluster identity $X = \frac{1}{2} \sum u_i^2$ for zero-mean root displacements gives the square-root scale $\sqrt{t}$; sums are linear. These are realized score paths, including the cone boundary. $\square$

The binary whole-model coefficient inverse and cluster estimates retained in the supporting volume verify these hypotheses on every compact full-rank fixed-pattern stratum. The paired-contrast model verifies them directly in its probability coordinates. Thus the two settings share the efficient discrimination interpretation. This identifies the intrinsic meaning of a variational ray while leaving separate the number of independent observations needed to estimate it.

APPENDIX A. REAL FACTORIZATION AND POSITIVE-TIME ACCESSIBILITY

We isolate two points at which complex or formal order calculations would not suffice. All analytic functions in this appendix are real analytic.

Lemma A.1 (Factors of a normal-crossings product). *In the local analytic ring at a point with coordinates $(r_1, \dots, r_s, z)$, suppose*

$$fg = u \prod_{i=1}^s r_i^{a_i}, \quad u(0) \neq 0.$$

Then $f = u_f r^b$ and $g = u_g r^c$ with analytic units u_f, u_g and $b + c = a$. If instead

$$\sum_{j=1}^M R_j^2 = ur^a,$$

then every a_i is even and $R_j = r^{a/2} V_j$ with analytic V_j and $\sum_j V_j^2 = u$. In particular the V_j have no common zero near the point.

Proof. Each coordinate ideal (r_i) is prime: its quotient is the integral domain of analytic germs in the remaining coordinates. An analytic germ divisible by r_i can be divided analytically, as seen by its convergent power series. Let b_i and c_i be the finite maximal orders of divisibility of f and g by r_i . The primality just noted gives $b_i + c_i = a_i$. Divide by all these coordinate powers. The remaining factors have product u , so their values at the point are nonzero and each is a unit. This proves the first assertion without an unstated factorization assumption.

For the second, let b_i be the smallest r_i -divisibility order among the R_j . After dividing by $r_i^{b_i}$, at least one restricted germ on $r_i = 0$ is nonzero. A sum of squares of real analytic germs is identically zero only if every germ is zero: evaluate on a real neighbourhood. Thus the restricted sum of squares is nonzero and its r_i -order is exactly $2b_i$. It follows that $a_i = 2b_i$. Every R_j is divisible by every $r_i^{b_i}$, hence by their product, by successive coordinate division. Dividing the

identity proves the remaining assertions. Since u is nonvanishing and equals a real sum of squares, it is positive on a sufficiently small neighbourhood. $\square$

Lemma A.2 (Proper pruning and accessible faces). *Let Γ be compact subanalytic and equal to the closure of $\Gamma \cap \{t > 0\}$, where $t \geq 0$ is analytic on an ambient neighbourhood. Let $\psi : M \rightarrow \Gamma$ be a proper analytic surjection from a nonsingular real analytic manifold. Retain the closure in M of $\{t \circ \psi > 0\}$. Its image is still all of Γ . The same statement holds after a further proper analytic surjection onto M .*

Suppose after normal crossings a retained local orthant chart has

$$t \circ \psi = u(r, z)r^a, \quad r_i \geq 0, \quad u > 0,$$

and its full interior has positive time. For every coordinate face with $a_i > 0$, every choice of sufficiently small fixed positive values of the other r_j and an interior value of z gives a feasible positive-time analytic arc transverse to that face, by setting $r_i = s \downarrow 0$. Its time order is a_i . If $R \circ \psi = r^b V$ with V nowhere zero, its residual norm has order b_i on this arc.

Proof. For $y \in \Gamma$ choose a sequence $y_n \in \Gamma$ with $t(y_n) > 0$ and $y_n \rightarrow y$. Choose any lifts $x_n \in M$. Properness over the compact set Γ supplies a convergent subsequence with limit x , and continuity gives $\psi(x) = y$. Its limit is in the retained closure. This proves surjectivity after pruning; the same argument proves the assertion after any further proper surjection.

On a connected analytic component on which $t \circ \psi$ is not identically zero, its nonnegative analytic function has a zero set with empty interior, so the positive-time part is dense. Components on which it vanishes identically contribute no retained interior and may be discarded. On a local normal-crossings chart choose orthants whose interiors have positive time, replace their signed coordinates by nonnegative r_i , and take smaller closed boxes. The transverse arcs specified in the statement belong to those boxes and their images lie in Γ . Holding the other coordinates positive leaves a nonzero constant times s^{a_i} for time and a nonzero vector times s^{b_i} for the residual, proving both orders. This is an explicit feasible arc, not an appeal to a complex divisor. $\square$

Theorem A.3 (Proper real order presentation). *Let Γ be as in Lemma A.2, and let $R_1, \dots, R_M$ be analytic on an ambient neighbourhood. Suppose there is no common residual zero on $\Gamma \cap \{0 < t \leq t_0\}$. On a sufficiently small compact time neighbourhood of the zero-time fibre there is a finite proper-surjective system of real corner charts with*

$$t = u(r, z)r^a, \quad R = r^b V(r, z), \quad r_i \geq 0, \quad a_i = 0 \implies b_i = 0, \quad (\text{A.1})$$

where u and $|V|$ have positive lower and finite upper bounds. Every face used in a ratio b_i/a_i is reached by a feasible positive-time transverse arc. Assuming the zero-time fibre is nonempty, put

$$\rho = \max\{b_i/a_i : \text{retained chart faces with } a_i > 0\}.$$

Then, on the feasible germ, $|R| \geq ct^\rho$ for a constant $c > 0$, and an accessible arc attains order ratio ρ . If the positive-time slices are all nonempty for small time, their minimum residual norm has order exactly ρ .

Proof. The uniformization theorem of [3, Theorem 0.1] gives a proper analytic surjection from a nonsingular manifold onto the compact subanalytic set. Pull back t and R . Prune as in Lemma A.2. On each remaining component the analytic function $t \sum_j R_j^2$ is not identically zero, since the residual is nonzero at positive time. Apply the proper normal-crossings theorem [3, Corollary 4.9] to that function on a neighbourhood of the compact inverse image of the zero-time fibre. The resulting manifold and proper surjection preserve the feasible image. Prune once more.

In every resulting coordinate neighbourhood the product is a coordinate monomial times a unit. The first part of Lemma A.1 factors the time function and the sum of squares separately; its second part factors the vector residual and proves that its remaining vector factor is nowhere

zero. Shrink to finitely many closed corner boxes covering the compact inverse zero-time fibre. The positive-time orthants have $u > 0$, and the units have uniform bounds on these boxes. Their images cover the feasible germ: otherwise positive-time points with times tending to zero outside the image would have lifts converging, by properness, to the inverse zero-time fibre covered by the boxes. Choose the boxes with interiors covering that fibre to obtain the contradiction.

If $a_i = 0$ but $b_i > 0$, take $r_i = 0$ and all other divisor coordinates positive and sufficiently small. The time is positive whereas every residual coordinate vanishes, a contradiction. Thus (A.1) holds. Lemma A.2 supplies all the stated transverse arcs with the correct orders, including the face realizing the finite maximum.

Rescale the boxes so that $0 \leq r_i \leq 1$. The inequalities $b_i \leq \rho a_i$ imply $r^b \geq (r^a)^\rho$. Combining this with the unit bounds yields $|R| \geq ct^\rho$ on every chart and hence on the germ. On the maximizing transverse arc, t is a positive unit times s^{a_i} and $|R|$ is a positive unit times s^{b_i} . This gives an upper bound by a constant times t^ρ on all sufficiently small time slices, since the arc's time is eventually strictly monotone. The matching lower bound proves the final statement. $\square$

The theorem computes orders for an analytic residual on a subanalytic feasible graph. A continuous semialgebraic observation can be treated by taking its compact graph and using the analytic coordinate projections for time and residual. The polar-gap suprema in Proposition 2.4 likewise have semialgebraic graphs. Extracting their orders by these real charts and then testing the strict inequalities in (2.3) connects the resolution presentation to the finite-sheet theorem. The extra residues and limiting derivative matrices in (2.7) are indispensable; monomial orders alone are not asserted to recover them.

APPENDIX B. THE BINARY INVERSE: PRECISE QUANTIFIERS AND SCORE COLUMNS

We state explicitly the correction to the earlier coefficient-inverse argument. Let

$$K(z) = \alpha_1 f_1(z) u_1 v_1^\top + \alpha_2 f_2(z) u_2 v_2^\top,$$

where f_1, f_2 are coprime monic degree- d polynomials, $d \geq 1$, with roots in $[0, D]$, $\alpha_a > 0$, $\alpha_1 + \alpha_2 = 1$, and $U = (u_1, u_2), V = (v_1, v_2)$ are invertible strictly positive column-stochastic matrices. Let $T_1, \dots, T_\ell > D$ be distinct, with $\ell \geq 2d + 1$. Put $q(z) = \mathbf{1}^\top K(z) \mathbf{1}$ and $P_j = K(T_j)/q(T_j)$.

Lemma B.1 (Existence of a separating clock pair). *For each fixed clock index i there exists an index j such that*

$$\frac{f_1(T_j)}{f_1(T_i)} \neq \frac{f_2(T_j)}{f_2(T_i)}.$$

No assertion is made for an arbitrary prescribed pair. The matrix $K(T_j)K(T_i)^{-1}$ for such a pair has distinct eigenvalues and its eigenvectors recover the columns of U after normalization by their entry sums. On a compact full-rank fixed-pattern family a finite cover supplies locally chosen pairs with uniform eigenvalue gaps.

Proof. If the displayed equality held for every j , the degree-at-most- d polynomial $f_1 - f_1(T_i)f_2/f_2(T_i)$ would vanish at every clock, hence identically. Monicity would make $f_1 = f_2$, contradicting coprimality. A separating pair therefore exists. Matrix multiplication gives

$$K(T_j)K(T_i)^{-1} = U \operatorname{diag} \left(\frac{f_1(T_j)}{f_1(T_i)}, \frac{f_2(T_j)}{f_2(T_i)} \right) U^{-1}.$$

The two simple eigenlines and their stochastic normalizations are analytic locally. A chosen strict gap persists in a neighbourhood. Compactness supplies a finite cover by such neighbourhoods and the minimum of their finitely many positive gap bounds. This does not select one universal pair on the entire family. $\square$

Lemma B.2 (Displayed partial-fraction independence). *For distinct cluster locations r_j in one component, the first-order polynomial columns corresponding to cluster sums and repeated interior variances are, respectively,*

$$-\frac{f_a(z)}{z - r_j}, \quad -\frac{f_a(z)}{(z - r_j)^2},$$

with the second column retained only when the multiplicity is at least two and the cluster is interior. The inward sum at the upper endpoint reverses the sign of the first column. All these columns are linearly independent. Combined with the analytic coefficient/stochastic inverse, they imply injectivity of the full free-and-retained score matrix.

Proof. A linear dependence, divided by the nonzero polynomial f_a , is a rational identity

$$\sum_j \frac{a_j}{z - r_j} + \sum_{\substack{j \text{ repeated} \\ \text{and interior}}} \frac{b_j}{(z - r_j)^2} = 0.$$

At each repeated pole, multiplying by $(z - r_j)^2$ and evaluating there gives $b_j = 0$. Multiplying the remaining identity by $z - r_j$ and evaluating gives $a_j = 0$ for every cluster. A sign reversal does not affect independence. Different components have separately recovered polynomial coefficients, so their columns are independent in that product coordinate space.

For completeness, coefficient recovery from the normalized clock laws uses the monic normalizer $q = \mathbf{1}^\top K \mathbf{1}$. A homogeneous ambiguity has $\deg e < d$, $\deg H \leq d$, and $H(T_j) = e(T_j)P_j$. Thus the polynomial $qH - eK$ of degree at most $2d$ vanishes identically. Taking a marginal and multiplying by the inverse channel gives $q \mid ef_1$ and $q \mid ef_2$. A Bezout identity for the coprime components implies $q \mid e$, hence $e = 0$ and $H = 0$. A nonsingular minor gives the analytic coefficient inverse. Lemma B.1 recovers U . The row sums of $U^{-1}K(z)$ are $\alpha_a f_a(z)$; their leading coefficients give α_a , and division gives f_a . Normalizing these rows gives $v_a^\top$. Thus both channels, weights, and polynomial coefficients are recovered locally. Differentiating this inverse gives a left inverse in coefficient and stochastic coordinates. Applying it to a zero combined score reduces to the polynomial independence just proved and to zero stochastic coordinate increments. This proves the final assertion. $\square$

All compact-uniform statements here hold within the full-rank fixed-pattern set with the required gaps and probability margins. They do not assert uniformity through channel-rank loss, cluster merging, a vanishing design weight, or a discriminant not satisfying the polar-gap bounds.

APPENDIX C. RELATION TO THE PRESERVED MATHEMATICAL RECORD

The complete preceding principal article is retained as the supporting volume. Its whole-model local spectral theorem, binary rational-moment bounds, fixed-experiment two-variance theorem, finite affine realization, product construction, vector weighted cancellation, one-endpoint classification, and exact diagnostic examples are not replaced or removed. The present endpoint theorem extends the mixed-sign one-endpoint phenomenon to an open class in every endpoint dimension; it does not delete the earlier quadratic one-endpoint fibres.

The archive retains the full multiplicity and boundary atlas, the exact rank-deficient cubic fibre and identifying inequalities, the inverse through channel-rank loss, both cubic walls, signed second-order and critical equality-shell calculations, real weighted atlas, finite-jet and Puiseux-residue comparisons, and fixed-format entrance results. In particular, for the earlier cubic family $f = z(z - r)^2$, $g = (z - s)^3$, the previously established identifying locus remains, under its original hypotheses,

$$\{\det U \neq 0\} \cup \{(3D - r)s - 2rD < 0\} \cup \{(1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3 < 0\}.$$

The finite-map theorem is not substituted for this different model. The correction in Lemma B.1 is the operative clock-pair quantifier for the preserved inverse proof.

The assertions in the present article are proved here. Finite symbolic tests accompanying the revision check displayed algebra and representative active cones; they are not formal verification of the universal theorems. Likewise, source preservation and native compilation are evidence about the manuscript and its build, not substitutes for independent mathematical review.

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